**QUAID-I-AZAM UNIVERSITY ISLAMABAD**

**DEPARTMENT OF COMPUTER SCIENCE**




# DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

## SOFTWARE PROJECT TECHNICAL DOCUMENTATION


| NAME | FAIZAN AHMAD |
|---|---|
| REG# | 04072213038 |

| NAME | Zohaib Zafar |
|---|---|
| REG# | 04072312049 |


**Submitted to:**

**Dr. Umer Rashid**

# *Abstract*

## *Detection of clickbait through Machine Learning*

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

# Table Contents

**Chapter No. 01** ..... **Error! Bookmark not defined.**

**Software Project Management Plan** ..... **Error! Bookmark not defined.**

**1 Introduction** ..... **Error! Bookmark not defined.**

**1.1 Project Overview** ..... **Error! Bookmark not defined.**

**1.2 Project Deliverables** ..... **Error! Bookmark not defined.**

**2 Project Organization** ..... **Error! Bookmark not defined.**

**2.1 Software Process Model** ..... **Error! Bookmark not defined.**

**2.2 Roles and Responsibilities** ..... **Error! Bookmark not defined.**

**2.3 Tools and Techniques** ..... **Error! Bookmark not defined.**

**3 Project Management Plan** ..... **Error! Bookmark not defined.**

**3.1 Tasks** ..... **Error! Bookmark not defined.**

**3.1.1 Requirement Analysis Phase** ..... **Error! Bookmark not defined.**

**3.1.1.1 Requirements Identification** ..... **Error! Bookmark not defined.**

**3.1.1.2 Defining Use Cases** ..... **Error! Bookmark not defined.**

**3.1.1.3 Develop Domain Model** ..... **Error! Bookmark not defined.**

**3.1.1.4 ERD Diagram** ..... **Error! Bookmark not defined.**

**3.1.1.5 Software Requirements Specification Completion** ..... **Error! Bookmark not defined.**

**3.1.2 Design Phase** ..... **Error! Bookmark not defined.**

**3.1.2.1 Develop Design** ..... **Error! Bookmark not defined.**

**3.1.2.2 Developing Interfaces** ..... **Error! Bookmark not defined.**

**3.1.2.3 Class Diagram Designing** ..... **Error! Bookmark not defined.**

**3.1.2.4 Sequence Diagram Design** ..... **Error! Bookmark not defined.**

**3.1.2.5 Design Phase Verification** ..... **Error! Bookmark not defined.**

**3.1.2.6 Software Test Documentation** ..... **Error! Bookmark not defined.**

**3.1.3 Implementation Phase** ..... **Error! Bookmark not defined.**

**3.1.3.1 System Implementation** ..... **Error! Bookmark not defined.**

**3.2 Assignments** ..... **Error! Bookmark not defined.**

**3.3 Project Schedule** ..... **Error! Bookmark not defined.**

**3.4 Gantt Chart** ..... **Error! Bookmark not defined.**

**Chapter 2** ..... 6

**Software Requirement Specification** ..... 6

**2.1 Introduction** ..... **Error! Bookmark not defined.**

**2.2 Functional Requirements** ..... **Error! Bookmark not defined.**
**Use Case Diagram** ..... **Error! Bookmark not defined.**
***Use Case* Description** ..... **Error! Bookmark not defined.**
**UC-1 User Registration** ..... **Error! Bookmark not defined.**
**UC-2 User Login** ..... **Error! Bookmark not defined.**
**UC-3 Reset Password** ..... **Error! Bookmark not defined.**
**UC-4 User Logout** ..... **Error! Bookmark not defined.**
**UC-5 Input Headlines / News Text** ..... **Error! Bookmark not defined.**
**UC-6 Upload Dataset** ..... **Error! Bookmark not defined.**
**UC-7 Train Machine Learning Model** ..... **Error! Bookmark not defined.**
**UC-8 Detect Clickbait** ..... **Error! Bookmark not defined.**
**UC-9 View Prediction Result** ..... **Error! Bookmark not defined.**
**UC-10 Explain Prediction** ..... **Error! Bookmark not defined.**
**UC-11 View Model Accuracy and Performance** ..... **Error! Bookmark not defined.**
**UC-12 Save Detection History** ..... **Error! Bookmark not defined.**
**UC-13 View Detection History** ..... **Error! Bookmark not defined.**
**System Sequence Diagram** ..... **Error! Bookmark not defined.**
**SSD UC-1 User Registration** ..... **Error! Bookmark not defined.**
**SSD UC-4 User Logout** ..... **Error! Bookmark not defined.**
**SSD UC-6 Upload Dataset** ..... **Error! Bookmark not defined.**
**SSD UC-7 Train Machine Learning Model** ..... 7
**SSD UC-8 Detect Clickbait** ..... 7
**SSD UC-9 Explain Prediction** ..... 8
**SSD UC-10 Save Detection History** ..... 8
**SSD UC-11 View Detection History** ..... 9
**Domain Model** ..... 10
**Chapter 3** ..... **Error! Bookmark not defined.**
**Software Design and Description** ..... **Error! Bookmark not defined.**
**Sequence Diagrams** ..... **Error! Bookmark not defined.**
**SD UC-1 User Registration** ..... **Error! Bookmark not defined.**
**SD UC-2 User Login** ..... **Error! Bookmark not defined.**
**SD UC-3 Reset Password** ..... **Error! Bookmark not defined.**
**SD UC-4 Input Headline / News text** ..... **Error! Bookmark not defined.**

**SD UC-5 Text Preprocessing** ..... **Error! Bookmark not defined.**
**SD UC-7 Model Training** ..... **Error! Bookmark not defined.**
**SD UC-8 Clickbait Prediction** ..... **Error! Bookmark not defined.**
**SD UC-9 Explanation Generation** ..... **Error! Bookmark not defined.**
**SD UC-10 Save Detection History** ..... **Error! Bookmark not defined.**
**SD UC-11 View Detection History** ..... **Error! Bookmark not defined.**
**Activity Diagram** ..... **Error! Bookmark not defined.**
**Class Diagram** ..... **Error! Bookmark not defined.**
**Chapter No 4** ..... **Error! Bookmark not defined.**
**Software Testing** ..... **Error! Bookmark not defined.**
**Introduction** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-1 User Registration** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-2 User Login** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-3 Reset Password** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-4 User Logout** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-6 Upload Dataset** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-7 Train Machine Learning Model** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-8 Detection Clickbait** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-9 View Prediction Result** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-10 Explain Prediction** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-11 View Model Accuracy & Performance** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-12 Save Detection History** ..... **Error! Bookmark not defined.**
**Acceptance Test for UC-13 View Detection History** ..... **Error! Bookmark not defined.**
**Chapter No 5** ..... **Error! Bookmark not defined.**
**Conclusion and Future Enhancement** ..... **Error! Bookmark not defined.**
**Introduction** ..... **Error! Bookmark not defined.**
**Summary and Conclusion** ..... **Error! Bookmark not defined.**
**References** ..... **Error! Bookmark not defined.**

# Chapter 2

# Software Requirement Specification

## SSD UC-7 Train Machine Learning Model

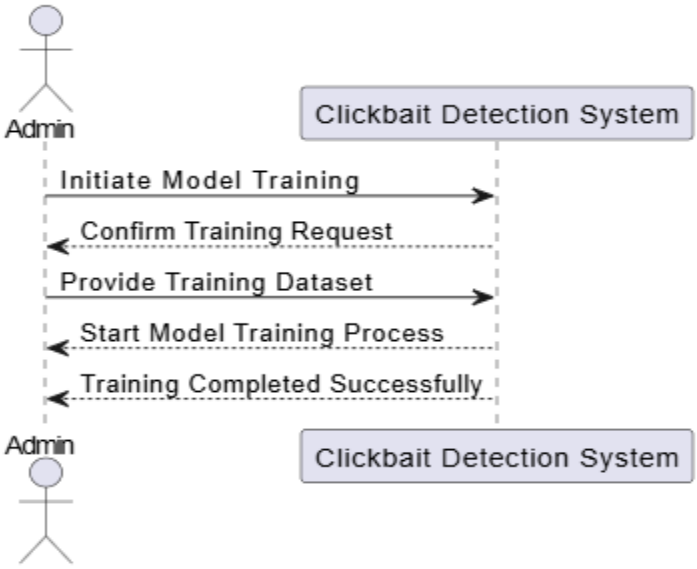



Figure 9 SSD UC-7 Train Machine Learning Model

## SSD UC-8 Detect Clickbait

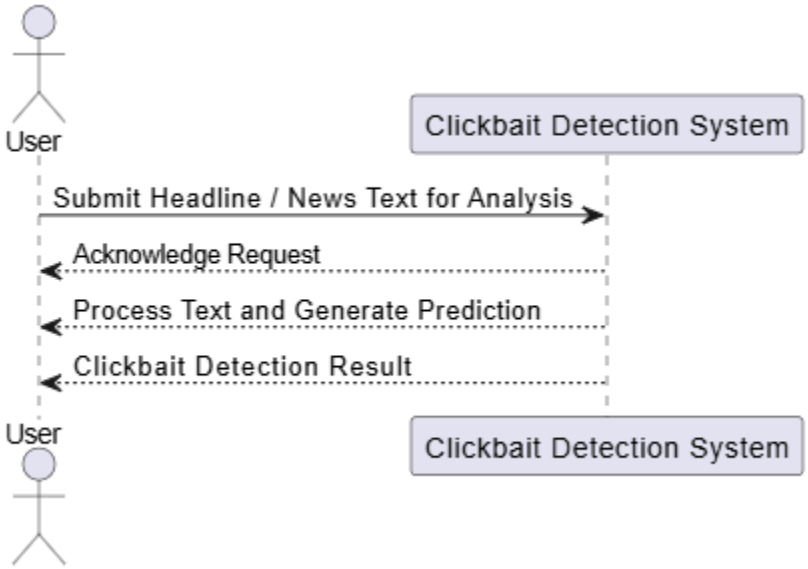



Figure 10 SSD UC-8 Detect Clickbait

## SSD UC-9 Explain Prediction

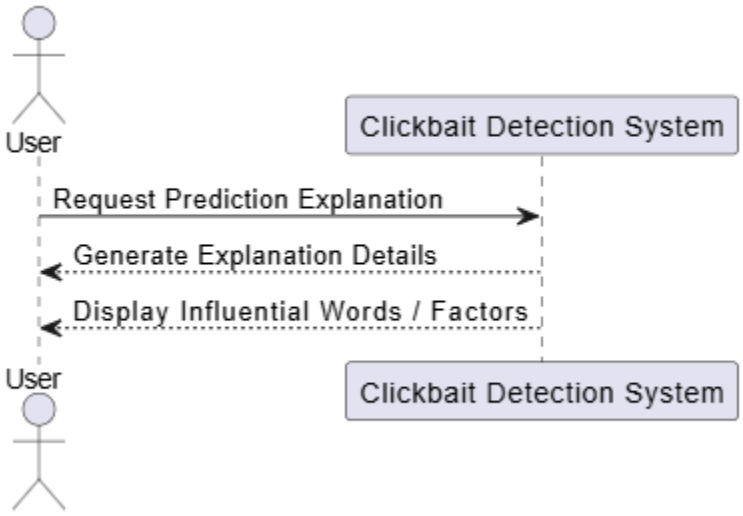



Figure 11 SSD UC-9 Explain Prediction

## SSD UC-10 Save Detection History

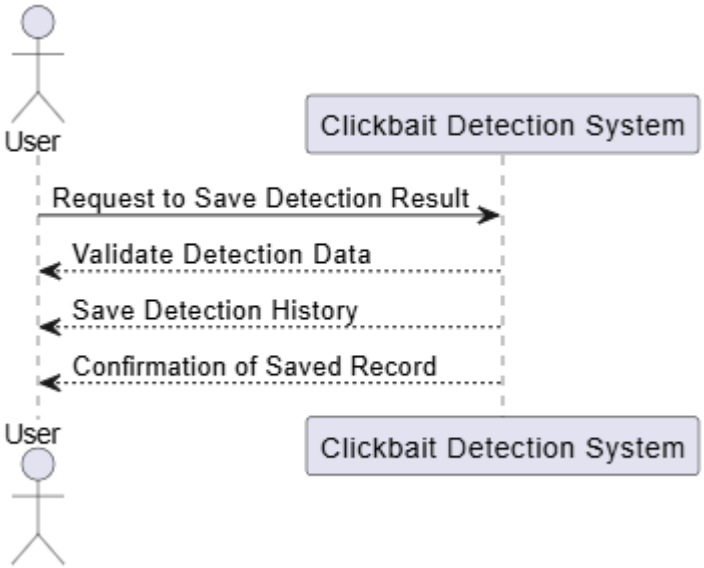



Figure 12 SSD UC-10 Save Detection History

**SSD UC-11 View Detection History**

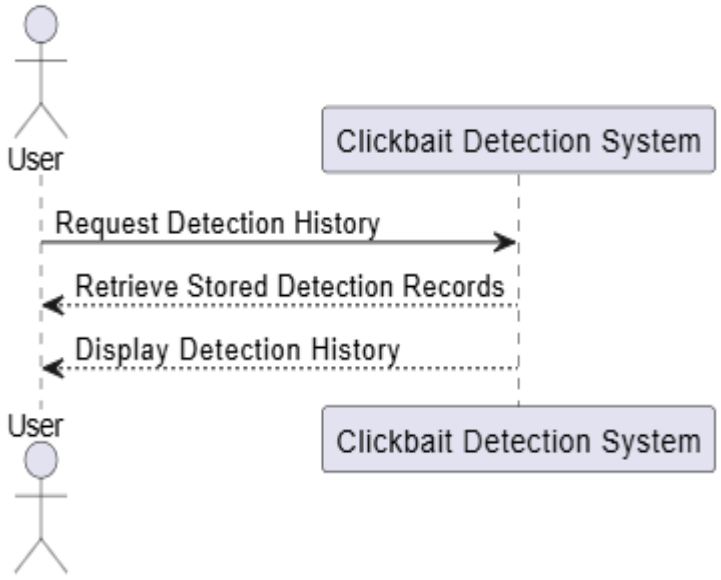



Figure 13 SSD UC-11 View Detection History

## Domain Model

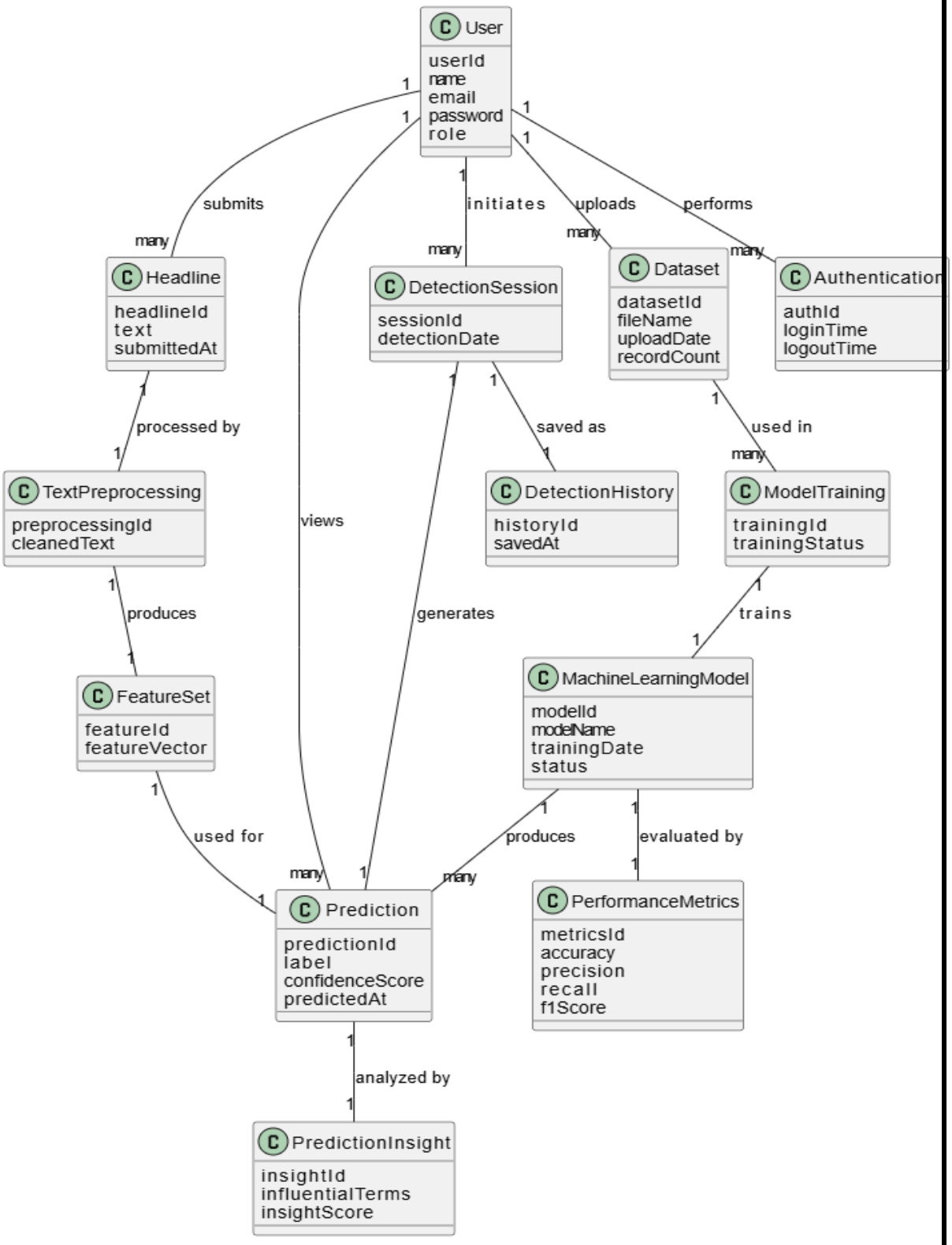



Figure 14 Domain Model